

# MMAE 350

## Homework 5: Nonlinear Position Analysis of an Offset Four-Bar Linkage

### Problem Description

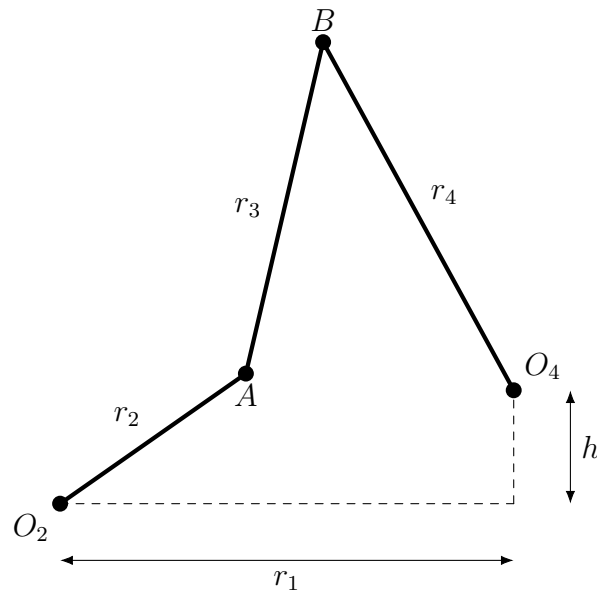
Consider the planar four-bar linkage shown below.

The fixed ground pivots are located at

$$O_2 = (0, 0), \quad O_4 = (r_1, h),$$

so the ground link has a horizontal length  $r_1$  and a vertical offset  $h$ .

The remaining link lengths are  $r_2$ ,  $r_3$ , and  $r_4$ . The input crank angle is  $\theta_2$ . The unknown angles are  $\theta_3$  and  $\theta_4$ .



Use the following parameter values:

$$r_1 = 4.0, \quad r_2 = 2.0, \quad r_3 = 3.0, \quad r_4 = 3.5, \quad h = 1.0.$$

## (a) Constraint Equations (6 pts)

Using vector addition and geometry, derive the two nonlinear loop-closure equations

$$f_1(\theta_3, \theta_4) = 0, \quad f_2(\theta_3, \theta_4) = 0,$$

that enforce closure of the offset four-bar linkage.

Write your result in vector form

$$\mathbf{F}(\theta_3, \theta_4) = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}.$$

Clearly indicate how the vertical offset  $h$  enters the equations.

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## (b) Newton Solver (8 pts)

- Derive the  $2 \times 2$  Jacobian matrix

$$\mathbf{J} = \frac{\partial \mathbf{F}}{\partial (\theta_3, \theta_4)}.$$

- Implement Newton's method to solve

$$\mathbf{F}(\theta_3, \theta_4) = 0$$

for a specified value of  $\theta_2$ .

- Use tolerance  $10^{-8}$ .
- Report the number of iterations required for convergence and the final residual norm.

Use the initial guess

$$\theta_3^{(0)} = \frac{\pi}{6}, \quad \theta_4^{(0)} = \frac{\pi}{3}.$$

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## (c) Motion Simulation (6 pts)

Let

$$\theta_2 \in \left[ \frac{\pi}{8}, \frac{\pi}{8} + 2\pi \right].$$

- Compute the linkage configuration at 50 equally spaced values of  $\theta_2$ .
- Use continuation (the converged solution at one step as the initial guess for the next).
- Generate an animation of the mechanism.